

Submission Instructions

This assignment consists of a single practical (coding) question, submitted once per group.

- Read the instructions carefully before starting the assignment.
- The practical part is submitted **once per group**: only one group member should submit it. Before submitting, one representative from each group must create a submission group on Moodle and add all group members to it (there is no need to change the Moodle submission group name).
- Name the ZIP file using your group serial number from the [Project Group Members](#) table:

`ex3p_<group_serial_number>.zip`

For example, if your group serial number is 53, the file is `ex3p_53.zip`. If your group does not appear in the table, please contact us immediately through the Private Forum.

- The ZIP must contain a `code/` folder and an `output/` folder:
 - `code/` — the Python code used to solve the assignment (if you work in a Jupyter notebook, place it here), and a `README.txt` with run instructions if needed.
 - `output/` — a consolidated report PDF that includes all plots, lists, examples, regular expressions, and explanations, plus any additional output files produced by the code.
- The consolidated report PDF must include the names and ID numbers of all group members and the name of the chosen book.
- All required outputs (plots, lists, examples, and analyses) must be included in the report, not only saved as files.
- Any plots must be informative and easy to read, with a title, axis labels, and a legend when needed.
- Follow the News forum and the HW forum for any updates or clarifications.
- Points may be reduced for submissions that do not follow the required file name, folder structure, or formatting.

Practical Part: Text Mining (Group Submission)

Download a book from Project Gutenberg <https://www.gutenberg.org>, preferably one that you know. You should submit two things: the code and the answers. You can use existing NLP and visualization packages (we recommend NLTK or spaCy for NLP). For the tag clouds, you can use online tools.

- (a) Write the name of the book you have chosen.
- (b) Tokenize the text. Count occurrences for each token. Plot the results (y axis: log frequency, x axis: log rank). Add the plot to the answers PDF.
- (c) Print a list of the top 20 tokens. Add the list to the answers PDF.
- (d) Repeat (b) after removing stopwords. Add the plot to the answers PDF.
- (e) Repeat (b) with stemmed text. Add the plot to the answers PDF. Add 3–4 sentences to the answers PDF explaining the difference in the count occurrences when looking at the original counts per token, after removing stopwords, and after stemming.
- (f) Show one example sentence where POS tagging made a mistake (explain). Add the example and explanation to the answers PDF.
- (g) Run POS tagging and create a tag cloud (word cloud) of proper nouns (NNP, NNPS). Does it correspond to what you know about the book? Add the tag cloud and explanation to the answers PDF.
- (h) Write a regular expression to find the set of all strings with two consecutive repeated words, with potential punctuation between them (e.g., “well well”, “so so”, and “no, no”). (Hint: `\b` is a word boundary, and `\1` references the first captured match.) Run it on your text and report any matches found in the answers PDF. Add the regular expression to the answers PDF as well.

Good Luck!